

Database Systems

Anomalies

Anomalies in Relational Model



Anomalies in the relational model refer to inconsistencies or errors that can arise when working with relational databases, specifically in the context of data insertion, deletion, and modification. There are different types of anomalies that can occur in referencing and referenced relations which can be discussed as:

These anomalies can be categorized into three types:

- Insertion Anomalies
- Deletion Anomalies
- Update Anomalies

Anomalies in Relational Model



- **Insertion Anomalies:** These anomalies occur when it is not possible to insert data into a database because the required fields are missing or because the data is incomplete. For example, if a database requires that every record has a [primary key](#), but no value is provided for a particular record, it cannot be inserted into the [database](#).
- **Deletion anomalies:** These anomalies occur when deleting a record from a database and can result in the unintentional loss of data. For example, if a database contains information about customers and orders, deleting a customer record may also delete all the orders associated with that customer.
- **Update anomalies:** These anomalies occur when modifying data in a database and can result in inconsistencies or errors. For example, if a database contains information about employees and their salaries, updating an employee's salary in one record but not in all related records could lead to incorrect calculations and reporting.

Delete Anomalies



<u>Student ID</u>	Student Name	Activity	Fee
9901	Billal	Basketball	200
9902	Shahzad	Football	300
9922	Sarfaraz	Cricket	500
9811	Parvez	Football	300

- What information do we lose if Binay quits Basketball?
- We would lose the price of 'Basketball'.
- This is the deletion anomaly that occur when relations are not fully normalized.
- When you delete some information and lose valuable related information at the same time.

Insert Anomalies



- If we want to record a new activity, but no one has yet taken it. Can we insert this information?
- We cannot do so; we need a student ID because the student ID is part of the primary key and therefore cannot be null. This is an insert anomaly.

<u>Student ID</u>	Student Name	Activity	Fee
9901	Billal	Basketball	200
9902	Shahzad	Football	300
9922	Sarfaraz	Cricket	500
9811	Parvez	Football	300

Update Anomalies



<u>Student ID</u>	Student Name	Activity	Fee
9901	Billal	Basketball	200
9902	Shahzad	Football	300
9922	Sarfaraz	Cricket	500
9811	Parvez	Football	300

- If we wanted to change the cost of football to '500', we would have to do it for every tuple where someone was playing football .
- Any change made to your data will require you to scan all records to make the change. This is called the update anomaly.

Removal of Anomalies



- These anomalies can be avoided or minimized by designing databases that adhere to the principles of normalization. Normalization involves organizing data into tables and applying rules to ensure data is stored in a consistent and efficient manner. By reducing data redundancy and ensuring data integrity, normalization helps to eliminate anomalies and improve the overall quality of the database
- According to **E.F.Codd**, who is the inventor of the [Relational Database](#), the goals of Normalization include:
 - It helps in vacating all the repeated data from the database.
 - It helps in removing undesirable deletion, insertion, and update anomalies.
 - It helps in making a proper and useful relationship between tables.

Disadvantages of Anomalies in Relational Model



- **Redundancy:** When the same data is stored in various locations, a relational architecture may cause data redundancy. This can result in inefficiencies and even inconsistent data.
- **Complexity:** Establishing and keeping up a relational database calls for specific knowledge and abilities and can be difficult and time-consuming.
- **Performance:** Because more tables must be joined in order to access information, performance may degrade as a database gets larger.
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